

Mid2 Solution - CN Sp24

Q1:

[3+4+4 = 10 Marks]

- a) List down any three responsibilities of the rdt_send() i.e reliable data send function in reliable data transfer protocol.
i. timeout management, ii. packet transmission, iii. sequence number assignment, iv. error detection, v. packet resend
- b) Checksum is an error-detecting code used in many Internet standard protocols, including IP, TCP, and UDP. Following data blocks are received with Checksum= **0001111001101110**. You need to verify the integrity of the received data using the given checksum.

0111001010110011	1011001110101000	1011101100110111
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- c) A computer with IP Address: 192.28.33.21 sends a request to another computer. The request contains following UDP header: **0050CB84001C001C**. Categorize content of above UDP header by answering the following questions
- What is the source port number?
 - What is the socket address of the sender end?
 - What is the total length of the user datagram?
 - What is the length of data?

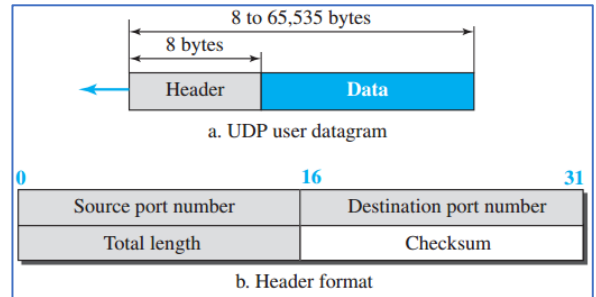


Figure 1. UDP header and payload format

Solution:

(b)

(c)

1st 0111001010110011
 2nd 1011001110101000

0010011001011011

0010011001011011

1011101100110111

1110000110010011

Receiver Calculated Checksum → 0001111001101110

0001111001101110

After Comparing one bit ERROR

a) Source Port no = 0050
 $= 0 \times 16^3 + 0 \times 16^2 + 5 \times 16^1 + 0 \times 16^0$
 $= 80$

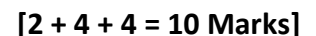
b) Socket Address = 192.28.33.21 | 80

c) Total length = 001C
 $= 0 \times 16^3 + 0 \times 16^2 + 1 \times 16^1 + 12 \times 16^0$
 $= 16 + 12$
 $= 28 \text{ bytes}$

d) Length of Data = 28 - 8 = 20 bytes (Total 16)

[5+5 = 10 Marks]

- (c)



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